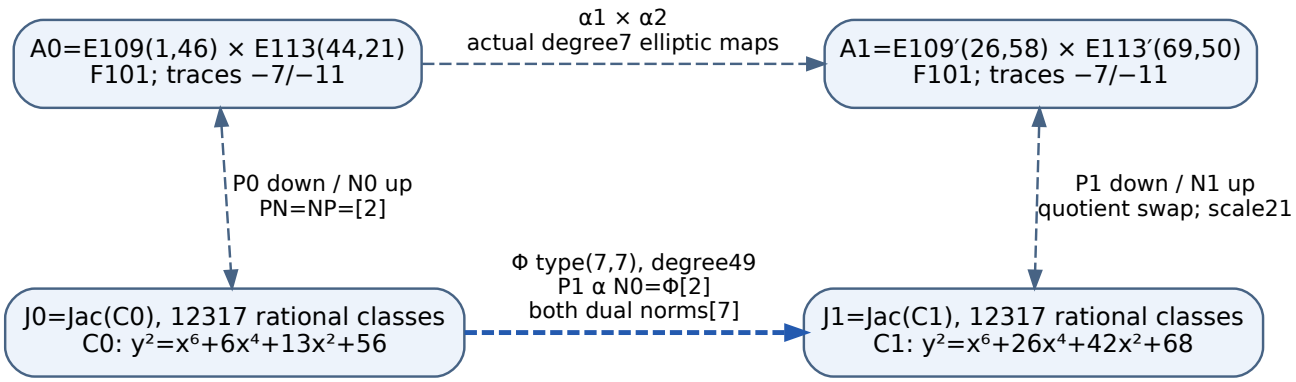
A fixed F101 degree 49 Jacobian transfer

2026-11-11 · Chained ECDLP Transfers · CT20

Unsigned scientific standing. Actual Phi/PhiHat evaluate both full rational groups,with a written global kernel-factor proof. Both signed dual norms[7], bijections,paired additions and every original109 source scalar pass. PCC6a1231/c78fb3. This is an actual same-prime map,not a solver benchmark. [pcc:unverified]



Full geometric2-kernel:16 points; norm graph4/kernel4
Both rational domains:12317-class dual/bijection checks
All109 original source scalars; N1 Φ f*P=([2]α1(P),0)
Written global proof + partial Lean helpers; scientific standing unsigned

Fixed curves and actual quotient matching

C0:$y^2=x^6+6x^4+13x^2+56$;C1:$y^2=x^6+26x^4+42x^2+68$. Both have12317=109· 113 rational Jacobian classes. Selected elliptic7-kernels[72,100,35,1] and [59,46,8,1] map(1,46)/(44,21) to(26,58)/(69,50). The C1 quotient order is swapped,with shifts76/14 and coordinate scale21. Actual coefficient equations and complete point-set bijections are checked. [pcc:unverified]

Model inventory corroborates cryptoPR3920/sourcecc3229d. Predeclared lexical kernel order selects raw[19,0,86,0,34,0,79],w=33Y. The other supported monic model is[33,0,44,0,92,0,1];raw leading41/46 remain unsupported for direct monic normalization. No silent twist or new discovery claim. [pcc:unverified]

The missing global bridge is supplied in writing

Actual norm/pull pairs satisfy PN=NP=[2]. A=P1 alpha N0 has degree 784. Its norm graph on ALL16 geometric J0[2] classes has image4/kernel4. Actual paired image roots satisfy UV=68,so both quadratic pullbacks agree and their2-torsion sum is zero. The entire finite-etale J0[2] is killed. The quotient universal property gives A=Phi[2],degree 49,defined overF101. Adjoints give[4]PhiHat Phi=[28];torsion-free HOM cancellation yields both dual norms[7]. Point groups are not assumed torsion-free. [pcc:unverified]

Evidence,formal scope and next chain

Complete bounded correctness

Both source/target12317-class corpora have exact signed dual[7],image-set bijections and12317 deterministic paired-addition cases per direction. These are not all12317² addition pairs;global additivity follows from the morphism construction. All 109 source scalars/nonzero image and direct target coordinates([2]alpha1(P),0) pass. Source norm[14] recovers with 39 mod109;Jacobian inverse7 is78. [pcc:unverified]

All 444 old/new elliptic-point extension pulls agree with the existing reciprocal/balanced engine,including4 origins,4 ramified U0 and4 V0 infinity fibers. Existing FpR/Kohel,balanced and tower engines remain unchanged;native/separate102/five-input guards remain distinct at75a737b8. The earlier independent Montgomery replay covers elliptic kernels/maps; this new whole-Jacobian evaluator is not a second signed verdict. [pcc:unverified]

Genuine F101 Jacobian distinction

Each elliptic pull kernel lies in E[2];all three nonzero2-torsion images are nonzero/distinct. Hence the factors embed. Different traces−7/−11 forbid cross homomorphisms,so each trace-type elliptic subvariety is unique. An unpolarized F101-Jacobian isomorphism would force E1≅E1′,contradicted by j12≠87. The full written proof is crypto:SEVEN_JACOBIAN_PROOF.md. No algebraic-closure distinction is claimed. [pcc:unverified]

Faithful partial formalization

Eight SevenJacobian.lean lemmas pass gate8/local bindingPCCb320bb: matched supports,kernel fiber constancy,unique functional descent/additivity, conditional Hom cancellation and source recoveries. Project total80, 0 trusted native uses. Actual finite fields,Picard maps,regular quotients, polarizations and elliptic-factor uniqueness remain written geometry. No weaker helper is presented as the full theorem. [pcc:unverified]

IDC1hef947795d0f8d235 and IDC1he970a46be821e787 carry exact Lean-source pointers and native32+16-point checks over2^61−1 for scaling/support polynomials. The checker does not execute Lean or express Hom composition. RootPCC3ed6c9. Two failed Lean attempts and the corrected Clippy iterator diagnostic are retained;no assumption,gate or arithmetic engine weakened. [pcc:unverified]

Continuing work and sources

Next:64 product4-torsion lifts give Phi on J0[2],then a transported Richelot kernel/R1 and a globally factored odd Psi from the accepted six-cycle target. That chain is not yet implemented by this checkpoint. Direct elliptic/source cost controls remain required;S3/live-budget still block benchmarks/solvers/paid work. Root ledger/archive and KB/vault gaps persist. [pcc:unverified]

Read HLP arXiv math/9809210v1 Section3.2 Propositions3/4/proof and Milne Abelian Varieties v2.0(2008),Remark8.12. Full source hashes,proof map, records,commands and limits:SEVEN_TRANSFER.md and the root proof report. No novelty or cross-prime transfer claim. [pcc:unverified]